(KTXFI1EBNF) Physics I. Homework No. 4

May 23, 2026

1. Given the sum of two one-directional oscillation as the following $x(t) = 4\sin(\omega t) + 5\cos(\omega t)$. This describes the displacement - time function of the oscillation, where t represents time, ω denotes the circular frequency of the oscillation, and ω is a constant.
   a) Prove that the above x(t) oscillation, given by the displacement-time function, performs simple harmonic motion.
   b) What is the value of the resultant oscillation given by the above displacement-time function?
2. Consider three one-directional oscillations with equal frequencies, whose amplitudes and initial phases are given as follows:

$$\begin{aligned} A_1 &= 5\cdot 10^{-3}, \mathrm{m} \quad , \quad \varphi_1 = \pi/5 \\ A_2 &= 2\cdot 10^{-3}\,\mathrm{m} \quad , \quad \varphi_2 = \pi/4 \\ A_3 &= 2\cdot 10^{-3}\,\mathrm{m} \quad , \quad \varphi_3 = 13\pi/11 \end{aligned}$$

   Let's combine these three oscillations. Calculate the amplitude and initial phase of the resultant vibration!
3. Calculate and determine the curve described by the resultant of the perpendicular vibrations given by the equations:

$$\begin{aligned} x(t) &= x_0\cos(\omega t), \\ y(t) &= y_0\sin(\omega t). \end{aligned}$$

4. Let $f(\omega t \pm kx)$ be a function whose argument is at least twice differentiable, continuous, and arbitrary, where ω: constant circular frequency, k : wavenumber (constant), φ initial phase (constant), t : time, x spatial coordinate. Prove that the function f described above is a solution to the one-dimensional wave equation.
5. The displacement of a wave as a function of time is given by

$$y(x,t) = 0.27\sin(12x - 500t)\,mm,$$

   where x is in meters and t is given in seconds. Determine at the point x = 0.2 meters in the case of a transverse wave:
   a) its velocity,
   b) its acceleration!
   c) What is the displacement of the point at t = 4s seconds?

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